

Do Celestial Bodies Influence Tropical Cyclone Behaviour?

A Rigorous Exploratory Analysis of Vedic Astronomical and Panchang Features

Against a Decade of Global Tropical Cyclone Track Data

Anonymous Submission

Empirical investigation — not intended as endorsement of astrological claims

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Abstract

We present a large-scale exploratory data analysis examining whether features derived from Vedic Panchang (Hindu lunar calendar) and classical astronomical ephemeris calculations carry statistically significant associations with tropical cyclone intensity and behaviour. Our dataset spans 944 tropical cyclones observed between July 2016 and January 2026, producing 47 533 track observations enriched with 282 derived features covering standard meteorological quantities, Skyfield-computed planetary positions, and Vedic categorical variables (Tithi, Nakshatra, Yoga, Karana, Paksha, Samvatsara, etc.).

We apply a seven-stage analysis pipeline: (1) data transformation and feature engineering, (2) basic exploratory analysis with multiple-comparison correction, (3) advanced feature importance ranking with storm-grouped cross-validation, (4) SHAP/LIME explainability, (5) comprehensive lifecycle and circular-statistics analysis, (6) formal hypothesis testing, and (7) block-wise predictive modelling.

Key findings: (i) Adding Vedic features to a meteorological baseline *degrades* out-of-sample RMSE by 12.6% (baseline 4.04 kt vs. Vedic 4.55 kt), indicating that Vedic features add noise rather than predictive signal. (ii) A marginal association between lunar phase and rapid intensification ($\chi^2 = 19.66$, $p = 0.0064$, 7 df) survives Bonferroni correction ($\alpha_{\text{adj}} = 0.0071$), but the effect size is small and the physical mechanism is unclear. (iii) Samvatsara-year ANOVA reaches $p < 10^{-130}$, but this reflects known interannual climate variability (ENSO/PDO), not Hindu calendar cycles. (iv) Storm-grouped cross-validation reveals that all tested tree-ensemble models achieve negative R^2 (-0.09 to -0.11), confirming that the astronomical feature block provides no out-of-storm generalisation. (v) Geospatial and track-geometry features (*latitude*, *longitude*, *spline_bearing_deg*) remain the dominant predictors throughout, as expected from tropical cyclone climatology.

We discuss methodological pitfalls — cyclical-variable treatment, within-storm autocorrelation leakage, and the Samvatsara confound — and provide corrected analyses for each. Our conclusion is that the current evidence does not support a predictive or associative role for Vedic astronomical constructs in tropical cyclone intensity.

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1 Introduction

Tropical cyclones are among the most destructive natural phenomena on Earth, responsible for billions of dollars in damage and thousands of casualties annually. Accurate intensity prediction remains an open research challenge: despite advances in numerical weather prediction and statistical modelling, the 24–48 hour intensity forecast skill of state-of-the-art models plateaus because of complex multi-scale interactions involving sea surface temperature (SST), vertical wind shear, moisture fields, and internal dynamics [DeMaria et al., 2014, Zhang & Tao, 2018].

An independent thread of inquiry asks whether *astronomical* or *astrological* quantities — planetary positions, lunar phase, Vedic calendar cycles — are associated with cyclone behaviour. Such a hypothesis is physically speculative: gravity from the Moon and Sun produce tidal stresses on the ocean and atmosphere, but the magnitudes involved are many orders of magnitude below the energy budget of a mature tropical cyclone. Nevertheless, the hypothesis is empirically testable, and a rigorous null-result analysis can clarify the upper-bound size of any such associations, guiding future data collection.

This work assembles the first large-scale dataset merging tropical cyclone track archives with Vedic Panchang / Jyotisha-Vedanga astronomical quantities. We subject this dataset to a carefully controlled analysis pipeline with explicit multiple-comparison correction, storm-grouped cross-validation, and circular statistics for angular planetary variables. We report results transparently, including negative or null findings.

1.1 Research Questions

- RQ1.** Do Vedic Panchang features improve out-of-sample prediction of cyclone wind intensity beyond a meteorological baseline?
- RQ2.** Is there a statistically robust association between lunar phase and the rate of rapid intensification (RI) events?
- RQ3.** Do planetary speed signatures (direct vs. retrograde motion) correlate with cyclone trajectory dynamics?
- RQ4.** Does the Vedic Samvatsara year cycle explain variance in storm intensity beyond interannual meteorological variability?
- RQ5.** Which astronomical or Vedic features show the most consistent importance across multiple analysis methods (tree models, SHAP, LIME, mutual information)?

2 Data

2.1 Source Data

Cyclone track data. We use 944 JSON files from a proprietary storm archive covering the period 2016-07-15 to 2026-01-24 UTC. Each file describes a single tropical cyclone with timestamped position, wind speed (kt), minimum central pressure (hPa), storm type classification, and advisory metadata. After join and deduplication, the dataset contains 47 533 track observations representing 735 distinct storms (a subset of the 944 files that yielded usable track rows after merging).

Astronomical ephemeris data. For each track observation, geocentric and topocentric positions for nine solar-system bodies (Sun, Moon, Mercury, Venus, Mars, Jupiter, Saturn, Uranus, Neptune) are computed using the Skyfield Python library [Rhodes, 2019]. Per-body fields include right ascension, declination, ecliptic longitude and latitude, altitude and azimuth above the local horizon, and sin/cos-transformed variants.

Vedic Panchang data. Vedic calendar attributes are extracted from a Drig-Ganita (observational) Panchang computation engine stored alongside each track record. These include:

- **Tithi** (lunar day, 1–30), **Nakshatra** (lunar mansion, 1–27), **Yoga** (luni-solar yoga, 1–27), **Karana** (half-tithi, 1–11), **Paksha** (fortnight: Shukla/Krishna), **Moonsign** (Rashi, 1–12), **Sunsign** (solar rashi), **Drik Ritu** (season)
- **Samvatsara** (60-year Jupiter cycle), **Drik Ayana** (solstice half)
- Numeric Vedic planetary quantities: full degree (0–360°), padam (nakshatra sub-division), speed (degrees/day), right ascension, declination for eight classical Vedic grahas (Surya, Chandra, Mangal, Budha, Guru, Shukra, Shani, Rahu/Ketu).

Derived features. The transformation pipeline computes:

- Temporal: *hour_utc*, *day_of_week*, *day_of_year*, *month*
- Storm dynamics: *wind_change* (kt/step), *pressure_change* (hPa/step), *displacement* (degrees), *movement_speed* (deg/h)
- Spline geometry: smoothed forecast track bearing, curvature, length, and endpoint (*spline_bearing_deg*, *spline_curvature*, etc.)
- Day-length: *dinamana_minutes* (daylight duration), *ratrimana_minutes*

2.2 Dataset Summary

Table 1: Dataset overview after transformation and join.

Property	Value
Observations (rows)	47 533
Feature columns	282
Unique storms	735
Date range	2016-07-15 to 2026-01-24 UTC
Wind speed: mean / median (kt)	49.9 / 40.0
Wind speed: SD / range (kt)	27.4 / [0, 170]
Pressure: mean (hPa)	989.9 (SD = 19.5)
Latitude range (°)	[−44.3, 64.0]
Longitude range (°)	[−179.8, 256.3]
Astronomical (Skyfield) features	90
Vedic planetary features	75
Vedic categorical features	10
Meteorological features	14

2.3 Missing Data

Table 2 summarises columns with $> 5\%$ missing values. The *storm_active* flag (99.7% null) and cone forecast features ($> 99\%$ null) are essentially absent and are excluded from all models. Spline features ($\approx 15.5\%$ null) are retained with listwise deletion where used.

Table 2: Columns with more than 5% missing values.

Feature Group	Missing (%)
<i>storm_active</i> flag	99.7
Cone geometry features (7 columns)	≈ 99.0
Spline geometry features (7 columns)	≈ 15.5
Anomaly label / score	10.9

3 Analysis Pipeline

The analysis proceeds through seven modular stages implemented as independent Python scripts. Each stage was audited for methodological correctness and corrected where necessary; the corrections are described in Section 4.

1. **Data transformation** (`01_data_transform.py`): JSON ingestion, Skyfield ephemeris computation, Vedic quantity extraction, feature engineering.
2. **Basic EDA** (`02_basic_eda.py`): Descriptive statistics, Kruskal–Wallis tests with Benjamini–Hochberg FDR correction, correlation screening.
3. **Advanced EDA** (`03_advanced_eda.py`): PCA, tree-ensemble feature importance with storm-*grouped* cross-validation, mutual information, clustering, redundancy analysis.
4. **SHAP/LIME** (`04_shap_lime.py`): Global and local model explanation with consensus feature identification.
5. **Comprehensive EDA** (`05_comprehensive_eda.py`): Lifecycle analysis, circular-linear correlations for angular features, retrograde analysis with significance tests.
6. **Hypothesis testing** (`06_hypothesis_testing.py`): Formal statistical tests for Vedic predictive lift (H1), lunar-phase RI association (H2), planetary speed – trajectory coupling (H3), Samvatsara intensity analysis, Mars/Rahu/Ketu analysis.
7. **Predictive modelling** (`07_predictive_model_stack.py`): Block-wise Random Forest regression comparing meteorology-only, panchang-enhanced, and full feature stacks using temporal forward-chaining cross-validation.

4 Methodological Corrections

During the pipeline audit we identified and corrected five methodological issues. We document them here for full transparency.

4.1 Circular-Variable Treatment (E2)

Planetary ecliptic longitude and Vedic full-degree quantities are angular variables $\theta \in [0\text{ř}, 360\text{ř})$. Applying Pearson r directly treats the interval as linear and incorrectly penalises the $0^\circ/360^\circ$ discontinuity (e.g., $\theta = 359\text{ř}$ is treated as maximally far from $\theta = 1\text{ř}$). We corrected this by replacing Pearson r with a circular–linear decomposition:

$$r_{\sin} = \text{Corr}(\sin(\theta), y), \quad r_{\cos} = \text{Corr}(\cos(\theta), y), \quad r_{\text{circ}} = \max(|r_{\sin}|, |r_{\cos}|) \quad (1)$$

Results reported in `vedic_degree_correlations.csv` now use r_{circ} .

4.2 Within-Storm Cross-Validation Leakage (C3)

The original 5-fold CV split rows randomly, allowing observations from the same storm to appear in both training and test folds. Because consecutive track points of a storm are strongly autocorrelated, this inflates R^2 estimates by effectively testing memorisation rather than generalisation. We replaced standard `KFold` with `GroupKFold(n_splits=5)` grouped by `storm_id`, ensuring all observations from a given storm appear in exactly one fold.

4.3 Multiple-Comparison Correction (B1)

The Kruskal–Wallis tests in Stage 2 test 10 Vedic category variables simultaneously. At $\alpha = 0.05$, approximately 0.5 false discoveries are expected by chance. We applied Benjamini–Hochberg (BH) FDR correction and added columns `p_adjusted_bh` and `significant_after_bh_correction` to `kruskal_wallis_wind.csv`. All conclusions use BH-adjusted p -values.

4.4 R^2 Discrepancy Between Stages 3 and 4

Stage 3 (grouped CV) reports $R^2 \approx -0.10$, while Stage 4 (SHAP on XGBoost) reports test $R^2 = 0.71$. The discrepancy arises from three sources: (1) Stage 4 uses a stricter $> 80\%$ completeness filter retaining only the most complete features; (2) Stage 4 uses a random 80/20 split, permitting within-storm autocorrelation to inflate the estimate; (3) Stage 3 uses `GroupKFold`. The grouped-CV R^2 is the appropriate measure of out-of-storm generalisation.

4.5 Rahu / Ketu Retrograde Independence

Rahu (North Lunar Node) and Ketu (South Lunar Node) are always exactly antipodal on the ecliptic and always move together at identical speed magnitude (retrograde $\approx -0.053\text{ř/day}$). Any statistical test run on both independently will produce *identical* results by astronomical definition; this is not a copy-paste error but an inherent constraint. Results for Ketu are labelled accordingly.

5 Results

5.1 Descriptive Statistics and Data Quality

Wind speed follows a right-skewed distribution (mean = 49.9 kt, median = 40.0 kt, SD = 27.4 kt, skewness = 1.35, range = [0, 170] kt). Central pressure is approximately symmetric (mean = 989.9 hPa, SD = 19.5 hPa, range = [882, 1019] hPa). Wind and pressure show a strong negative correlation ($r^2 = 0.886$).

The 208 numeric features can be grouped as: 90 Skyfield astronomical features, 75 Vedic planetary features, and a smaller meteorological/derived set. PCA on the 90-dimensional Skyfield feature matrix shows that 29 principal components are needed to explain $\geq 90\%$ of variance, confirming moderate redundancy but non-trivial intrinsic dimensionality consistent with the planet-body structure of the feature space (PC1 is dominated by Jupiter–Uranus–Neptune ecliptic longitude cluster; PC2 captures Sun/inner-planet altitude/azimuth patterns). A total of 239 feature pairs exhibit $|r| > 0.95$, predominantly among the sin/cos transforms of the same body position.

5.2 Kruskal–Wallis Tests on Vedic Categories (BH-Corrected)

Table 3 reports Kruskal–Wallis H statistics and BH-corrected p -values for ten Vedic categorical variables against wind speed. Eight of ten categories are significant after correction, with Drik Ritu (season: $H = 1214$) and Sunsign (solar zodiac: $H = 1123$) showing the largest effect. *However*, as noted in Section 6.1, Ritu and Sunsign are essentially time-of-year proxies that encode the annual tropical cyclone season cycle. Weekday is not significant ($H = 11.6$, $p = 0.072$).

Table 3: Kruskal–Wallis tests (wind speed by Vedic category) with BH correction. “Sig.” = significant at BH-adjusted $p < 0.05$.

Category	H	p (raw)	p (BH adj.)	Sig.
Drik Ritu (season)	1214.0	2.7×10^{-260}	2.7×10^{-259}	Yes
Sunsign	1123.1	5.9×10^{-234}	2.9×10^{-233}	Yes
Yoga	143.3	3.4×10^{-18}	8.6×10^{-18}	Yes
Nakshatra	116.1	2.4×10^{-13}	4.7×10^{-13}	Yes
Drik Ayana	102.6	4.0×10^{-24}	1.3×10^{-23}	Yes
Moonsign	77.9	3.7×10^{-12}	6.2×10^{-12}	Yes
Tithi	51.3	7.5×10^{-6}	9.4×10^{-6}	Yes
Karana	42.5	6.2×10^{-6}	8.8×10^{-6}	Yes
Paksha (fortnight)	8.1	4.4×10^{-3}	4.9×10^{-3}	Yes
Weekday	11.6	7.2×10^{-2}	7.2×10^{-2}	No

5.3 Feature Importance under Storm-Grouped Cross-Validation

Using `GroupKFold` (groups = storm_id), the RF, XGBoost, and LightGBM models trained on the full 186-feature set all show negative grouped CV R^2 :

Model	Grouped CV R^2	(SD)
Random Forest	−0.0901	±0.0397
XGBoost	−0.1089	±0.0500
LightGBM	−0.1010	±0.0262

Negative R^2 means the model performs worse than predicting the global mean wind for every new (unseen) storm. Despite this, model-internal feature importances indicate which features the model *uses* (not which are predictive in a generalisable sense). The top 10 averaged importances are shown in Table 4.

Table 4: Top-10 averaged feature importances (RF/XGB/LGB mean importance score). Note: models show negative grouped CV R^2 ; rankings are exploratory, not predictive.

Feature	Mean importance
<i>longitude</i>	0.0588
<i>latitude</i>	0.0420
<i>spline_end_lon</i>	0.0340
<i>dinamana_minutes</i>	0.0207
<i>vedic_chandra_speed_deg/day</i>	0.0203
<i>spline_bearing_deg</i>	0.0187
<i>elevation</i>	0.0179
<i>ratrimana_minutes</i>	0.0174
<i>mars_ecliptic_lat_deg</i>	0.0168
<i>sun_dec_deg</i>	0.0162

5.3.1 Caution on Spline Features

The features *spline_end_lon*, *spline_bearing_deg*, *spline_curvature*, *spline_displacement*, and *spline_length_deg* encode the smoothed forecast track geometry. Using these to predict current or next-step wind introduces potential *implicit leakage*: knowing the future trajectory implies knowing the future position of the storm. These features rank highly in importance but are excluded from the scientific (astronomical) block in the predictive model to avoid contaminating the comparison.

5.4 SHAP / LIME Explainability

The SHAP analysis uses an XGBoost model trained on features with > 80% completeness (186 features) with a random 80/20 split (train $R^2 = 0.784$, test $R^2 = 0.707$). **Note:** This R^2 is inflated relative to the grouped-CV estimate (≈ -0.10) due to within-storm autocorrelation in the random split.

Fourteen features appear in the top-30 of *both* SHAP and LIME importance rankings (consensus set):

- Geospatial: *latitude*, *longitude*, *nearest_lat*
- Track geometry (leakage risk): *spline_bearing_deg*, *spline_end_lon*
- Day-length: *dinamana_minutes*, *ratrimana_minutes*

- Solar: *sun_dec_deg*
- Planetary: *mercury_dec_deg*, *neptune_ecliptic_lon_deg*
- Vedic planetary speeds/degrees: *vedic_arun_speed_deg/day* (Uranus), *vedic_budha_full_degree* (Mercury), *vedic_budha_speed_deg/day*, *vedic_guru_full_degree* (Jupiter)

After removing leakage-risk spline features and day-length features (which encode latitude-driven seasonality), the robust non-redundant signals are *latitude*, *longitude*, and several planetary position quantities. Planetary quantities with high individual SHAP importance (e.g. *vedic_chandra_speed_deg/day*, *mars_ecliptic_lat_deg*) likely reflect a confound: planetary positions are correlated with calendar time and thus with the tropical cyclone season.

5.5 Hypothesis Testing Results

5.5.1 H1: Vedic Predictive Lift

We trained a Random Forest model to predict *wind_change* using: (a) meteorological features only (baseline), and (b) meteorological + Vedic features.

Model	RMSE (kt)	Change
Meteorological baseline	4.04	—
Meteorological + Vedic	4.55	+12.6% (worse)

Conclusion (H1): Vedic features degrade prediction. The null hypothesis that Vedic features have no predictive lift is *not* rejected; Vedic features actively harm out-of-sample performance, consistent with overfitting noise.

5.5.2 H2: Lunar Phase and Rapid Intensification

We define Rapid Intensification (RI) as a wind increase of ≥ 30 kt within a 24-hour rolling window (note: WMO standard is ≥ 35 kt; we document this discrepancy). With this threshold, 2 020 RI events were identified across 47 533 observations.

A chi-square test of the contingency table (lunar phase $\times$ RI/non-RI):

$$\chi^2 = 19.66, \quad \text{df} = 7, \quad p = 0.0064 \quad (2)$$

With Bonferroni correction across 7 main hypothesis groups ($\alpha_{\text{adj}} = 0.05/7 = 0.0071$), this result *barely* survives correction ($p = 0.0064 < 0.0071$).

The Waning Crescent phase shows the highest RI count (301 events vs. a cross-phase mean of ≈ 253). The effect size (Cramér’s $V \approx 0.02$) is small. **Conclusion (H2):** Marginal statistical evidence for a lunar-phase – RI association; physical interpretation requires caution.

5.5.3 H3: Planetary Speeds and Trajectory Dynamics

Spearman correlation was computed between 60 pairs of (planetary speed, storm trajectory feature). The top associations all involve outer-planet speed proxies (Jupiter, Saturn, Neptune) correlated with day-of-year encoded trajectory variables. Retrograde vs. direct

ANOVA showed no significant wind-change differences for any tested body after Bonferroni correction.

5.5.4 Samvatsara Analysis

The 60-year Vedic Samvatsara cycle maps, in the current dataset, to approximately one calendar year per Samvatsara (since ≈ 10 years of data are available, roughly 10 Samvatsara names appear). A one-way ANOVA of wind speed by Samvatsara reaches F -stat $\gg 10^{100}$, $p < 10^{-130}$.

Critical caveat: This test is mathematically equivalent to asking “does storm intensity differ across calendar years?” The answer is trivially yes due to interannual climate variability (ENSO, PDO, Atlantic Multidecadal Oscillation). The result carries no evidence for a Hindu calendar-cycle effect.

5.5.5 Mars / Rahu / Ketu Analysis

Analysis	Statistic	p -value
Mars+Rahu+Ketu model vs. baseline (RMSE)	+1.1% (worse)	—
Mars altitude band $\times$ RI (χ^2)	—	0.532
Retrograde vs. direct wind-change (Mars)	ANOVA F	0.916
Rahu retrograde vs. direct wind-change	ANOVA F	0.040
Rahu degree vs. wind-change (Spearman)	ρ	0.006
Ketu retrograde vs. direct wind-change	ANOVA F	0.040*

* Identical to Rahu by astronomical constraint (see Section 4).

Rahu degree Spearman $p = 0.006$ and retrograde ANOVA $p = 0.040$ fail Bonferroni correction ($\alpha_{\text{adj}} = 0.007$). Mars, Rahu, and Ketu features do not survive family-wise error rate control.

5.6 Block-Wise Predictive Modelling

Table 5 summarises the four feature blocks in temporal forward-chaining cross-validation (train / test split: 2016–2023 Oct / 2023–2026).

Table 5: Block-wise Random Forest regression results (predicting next-step wind). Lower RMSE is better.

Feature Block	Test RMSE (kt)	Test MAE (kt)
Meteorological only (14 features)	4.164	2.540
Stacked residual	4.193	2.583
Meteo + Panchang (18 added)	4.372	2.685
Full (all blocks, 190 added)	4.515	2.921

Key finding: The meteorological-only model achieves the best RMSE. Each additional block of astronomical or Vedic features increases RMSE, with the full model (+190 features) performing worst. This is consistent with the hypothesis testing result (H1) and confirms that the 193 astronomical features add noise.

RI classification note: The adaptive RI threshold selector for the classification task settled on ≥ 10 kt (next-step delta), which is fundamentally different from the 30 kt/24h rolling definition used in H2. Cross-script RI comparisons should not be made.

6 Discussion

6.1 The Seasonal Confound

The strongest Vedic categorical associations (Drik Ritu, Sunsign, Drik Ayana) all encode time of year: tropical cyclone seasons peak in summer–autumn months, so any calendar variable correlated with the season will show a wind association. This is not a Vedic astronomy effect; it is the tropical cyclone season signal re-expressed in calendar coordinates.

Similarly, Vedic planetary positions (full degree, speed) are functions of calendar time for the slow outer planets (Saturn, Jupiter, Rahu). Their apparent correlation with storm intensity is a proxy for year-level interannual variability.

6.2 Multicollinearity Between Vedic and Astronomical Features

Vedic planetary features (e.g. *vedic_budha_full_degree*) are computed from the same Skyfield ephemerides as the Skyfield features (e.g. *mercury_ecliptic_lon_deg*) but with different coordinate systems (sidereal vs. tropical ecliptic, offset by the Lahiri ayanamsha $\approx 23.9^\circ$). Any observed Vedic feature correlation is therefore geometrically equivalent to the corresponding Skyfield feature correlation. There are 239 feature pairs with $|r| > 0.95$; adding Vedic features on top of Skyfield features is nearly redundant and degrades models through noise amplification.

6.3 Variance Inflation Factors

VIF analysis on a representative 9-feature subset confirms low multicollinearity (all VIF ≤ 2.0 , well below the conventional danger thresholds of 5 and 10), suggesting the chosen meteorological predictors are not strongly collinear with each other. The problem is that the 193 astronomical features are collectively redundant *with each other*, not with the meteorological features.

6.4 Interpretation of SHAP / LIME Consensus Features

Of the 14 consensus features, 7 are straightforward seasonal or geographic signals: *latitude/longitude* (climatological intensity zone), *dinamana_minutes/ratrimana_minutes* (day length = latitude + season), *sun_dec_deg* (solar declination = season). Three are planetary quantities that, as argued above, likely capture seasonality. Only after removing these geographic and seasonal proxies do the remaining features (*vedic_budha_speed_deg/day*, *vedic_guru_full_degree*, *vedic_arun_speed_deg/day*) represent potential astrological signals; and even these are weak (importance < 0.02) in a model that generalises poorly across storms.

6.5 Limitations

1. **No meteorological driver features:** The dataset does not include SST, vertical wind shear, moisture, or upper-level flow, which are the primary determinants of tropical cyclone intensity. Absence of these features explains the negative grouped CV R^2 for all models.
2. **Coarse temporal resolution:** Panchang quantities and Skyfield ephemerides are computed at each track advisory time (≈ 6 -hourly), but cyclone intensity change operates on sub-6-hour scales during rapid intensification.
3. **Multiple basins:** The dataset covers all global basins simultaneously. Basin-specific effects (Atlantic AMO, Pacific ENSO sensitivity) are not separated, adding variance that confounds between-year comparisons.
4. **Spline leakage:** Spline track features may encode future trajectory information; their high importance scores should not be interpreted as evidence of out-of-sample predictive power.
5. **Single ML family:** Feature importance and SHAP are computed from tree-ensemble models only; linear and Gaussian process models might yield different feature rankings.

7 Conclusion

We conducted a rigorous, multi-method exploratory investigation into whether Vedic Panchang and astronomical ephemeris quantities carry statistically meaningful associations with tropical cyclone intensity and dynamics.

Our principal findings are:

1. Vedic features consistently *degrade* out-of-sample predictive performance relative to a meteorological-only baseline (+12.6% RMSE in H1, +8.5% RMSE in the block-wise model). There is no evidence of predictive lift.
2. The strongest Vedic category associations (Drik Ritu, Sunsign) are seasonal proxies; they encode the tropical cyclone season signal, not astrological effects.
3. A marginal lunar-phase – RI association ($p = 0.0064$) survives narrow Bonferroni correction but has Cramér’s $V \approx 0.02$, implying negligible practical importance.
4. All three tree-ensemble models achieve negative grouped-CV R^2 on the full feature set, confirming the astronomical feature block adds noise.
5. The dominant predictive features remain geospatial coordinates (*latitude*, *longitude*) and track-geometry quantities (*spline_bearing_deg*), consistent with established tropical cyclone climatology.

This null result has practical value: it defines the upper bound of any Vedic / Panchang contribution to cyclone intensity that can be detected in a decade-long global dataset at 6-hourly resolution. Future work should (a) include physical meteorological drivers (SST, wind shear), (b) focus basin-specific analyses, (c) apply gravitational tidal forcing models as a physically motivated replacement for empirical Vedic categories, and (d) explore

whether lunar-phase tidal signals appear more clearly in SST anomaly or mixed-layer depth data that preconditions the ocean for intensification.

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A Feature Block Definitions

Meteorological block (14 features) *wind, pressure, latitude, longitude, wind_change, pressure_change, displacement, movement_speed, hours_elapsed, spline_bearing_deg, spline_curvature, spline_displacement, spline_length_deg, elevation.*

Panchang block (18 features) Tithi (numeric), Nakshatra (numeric), Yoga (numeric), Karana (numeric), Paksha (binary), Moonsign (numeric), Sunsign (numeric), Drik Ritu (numeric), Drik Ayana (numeric), Samvatsara (numeric), *dinamana_minutes, ratrimana_minutes, day-of-week, day-of-year, month, hour_utc, lahiri_ayanamsha.*

Scientific block (190 features) All Skyfield per-body position features (RA/Dec, ecliptic lon/lat, altitude/azimuth, sin/cos transforms) plus Vedic planetary numeric quantities (full degree, padam, speed, RA, Dec). *Excludes spline_ prefix features to avoid leakage.*

B Circular-Linear Correlation Results (Top 10)

Vedic degree feature	$r_{\sin}$	$r_{\cos}$	r_{circ}
(see EDA/output/comprehensive/vedic_degree_correlations.csv for full table)			

All circular-linear correlations with wind speed are small ($|r_{\text{circ}}| < 0.15$ for all tested Vedic degree features), confirming that no angular Vedic planetary feature has a meaningful linear relationship with cyclone intensity.

C Pipeline Audit Summary

ID	Script	Issue	Severity
A2	06	Rahu = Ketu retrograde (astronomical constraint)	Critical
B1	02	No multiple-comparison correction (fixed: BH FDR)	Medium
C1	03	Negative CV R^2 ; feature ranks unreliable	Critical
C3	03	Same-storm CV leakage (fixed: GroupKFold)	Medium
D1	04	R^2 discrepancy vs. Stage 3 (documented)	Critical
E2	05	Circular degrees treated as linear (fixed: sin/cos)	Critical
E3	05	Retrograde no p -values (fixed: Mann–Whitney U)	Medium
F2	06	Vedic model worse than baseline (flagged in report)	High
F5	06	Samvatsara ANOVA = interannual confound (documented)	High
F6	06	No Bonferroni correction (fixed: $\alpha_{\text{adj}} = 0.007$)	Medium
G3	07	Spline feature leakage (removed from sci. block)	Critical
G2	07	Best model = meteo_only, not full (flagged in report)	High